

Ten percent human: an evidence-graded audit of the mTOR literature and the pathway's translational gap

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Abstract

The mTOR pathway is among the most heavily cited targets in ageing biology, and claims derived from it circulate widely in translational and public-facing contexts. What is rarely made explicit is the kind of evidence any given claim rests on. I report a systematic audit of a hand-curated corpus of 386 mTOR studies (1975–2026), in which every record carries an explicit classification of the biological system it was done in, assigned under a fixed rubric.

Three findings emerge. First, the human evidence base is thin: 39 of 386 studies (10.1%) are human work of any kind (systematic review, clinical trial, or observational study), while 305 (79.0%) are animal or mechanistic/in-vitro work. Second, the human evidence that does exist is overwhelmingly about disease indications rather than ageing: of 25 human clinical trials, 13 concern licensed uses (transplant immunosuppression, oncology, tuberous sclerosis complex / lymphangioleiomyomatosis) and two are physiology experiments in healthy volunteers, leaving ten ageing- or healthspan-oriented trials. Among those ten, every positive primary-endpoint result comes from an early-phase, small or uncontrolled design; the one adequately powered confirmatory trial (Mannick et al., 2021; $n = 1024$) missed its primary endpoint (26% vs 25%, $p = 0.65$) after a positive phase-2a by the same group; the only purpose-built healthspan RCT completed to date (PEARL; Moel et al., 2025; $n = 114$) reported a pre-registered null; and a pre-registered crossover trial in 63 healthy adults found that moderate dietary protein restriction did not raise autophagic flux at all (Singh et al., 2026), against a clear preclinical expectation that it should. Third, the proportion of human evidence is rising slowly but monotonically across eras — 0% before 2000, 5.5% in 2000–2009, 10.0% in 2010–2019, 14.7% from 2020 — meaning that even in the most recent period roughly six in seven curated studies remain pre-clinical.

I also report a structured gap layer derived from the corpus (ten hypotheses, each linked to a mean of 6.6 constituent studies), and describe a revision of the classification scheme itself prompted by external expert critique — a case in which the act of grading evidence exposed a defect in the grading rubric. The corpus, the rubric, and the generating code are openly available.

Keywords: mTOR; evidence grading; research synthesis; translational gap; meta-research; knowledge base

1. Introduction

The mechanistic target of rapamycin (mTOR) has been a central object of ageing research since rapamycin was shown to extend lifespan in genetically heterogeneous mice even when started late in life (Harrison et al., 2009). Three decades of work separate the identification of the mammalian target (Sabatini et al., 1994) from the current structural and nutrient-sensing picture (Saxton et al., 2015), and the pathway now anchors a large literature spanning biochemistry, model-organism longevity, oncology, and a small number of human trials.

That breadth creates a specific problem for anyone trying to read the field. A statement such as "mTOR inhibition improves immune function with age" is true of a phase-2a trial, was not confirmed in the corresponding phase-3, and is separately supported by mouse and cell work of an entirely different evidential character. In practice these strands are cited interchangeably. The claim travels; the system it was established in does not travel with it.

Most existing resources do not fix this, because they are not built to. Bibliographic databases index papers without classifying them. Pathway databases model molecular relationships without reference to the study designs underneath. Narrative reviews do the classification implicitly, in prose, in a form that cannot be queried or audited. Formal evidence-grading frameworks such as GRADE exist but are applied to specific clinical questions in systematic reviews, not maintained continuously across a whole pathway literature.

The Open mTOR Atlas was built to hold the missing layer: a curated corpus in which every study carries an explicit, rule-assigned label for the system it was done in, so that the composition of the evidence base itself becomes a queryable object. This paper is not a description of that resource. It is an audit of what the resource shows when you ask it about itself.

The question is deliberately narrow: **in a curated corpus of the mTOR literature, how much of the evidence actually speaks to humans, and how much of that speaks to ageing?**

2. Methods

2.1 Corpus construction

Studies were identified through PubMed searches across mTOR pathway biology, rapamycin and rapalog pharmacology, nutrient sensing, autophagy, and ageing-related outcomes, and were selected by hand rather than by an automated query. Selection favoured (a) primary discoveries that established a mechanism, (b) model-organism longevity and healthspan studies, (c) all identifiable human trials involving mTOR inhibitors or the dietary interventions that act through the pathway, and (d) papers reporting negative or contradictory results, which were sought deliberately because they are systematically under-represented in citation-driven discovery.

The corpus is therefore a curated collection, not a systematic review sample: it makes no claim to exhaustiveness, and the selection criteria, while stated, were applied by a single curator. Section 5 treats this as the principal limitation of the work.

At the census date (15 September 2026) the corpus contained **386 studies** spanning **1975–2026**. Of these, 381 (98.7%) are peer-reviewed journal articles; the remainder are preprints (2), a registered trial record (1), and two records with incomplete provenance metadata.

2.2 Classification by study system

Each study is labelled by the biological system the work was done in, assigned from its design rather than from its conclusions or its citation count. The classes occupied in the current corpus:

Class	Description	n	%
Systematic review	Synthesis of human data	2	0.5
Human trial	Interventional study in people	25	6.5
Human observational	Cohort, case-control, registry	12	3.1
Animal	Whole-organism model	107	27.7
Mechanistic	Cells, biochemistry, structure	198	51.3
Narrative review	Secondary synthesis, unsystematic	39	10.1
Preprint	Not yet peer reviewed	2	0.5
Registered trial	Protocol, no results yet	1	0.3

These classes are not a quality ranking, and the published resource deliberately does not display them in a ranked order. A mechanistic paper is not "worse" than a clinical trial. It answers a different question and licenses different claims, and a rigorous biochemical result can be far more reliable than an underpowered trial. The label says which system a result came from, not how much it counts. The reasoning behind that commitment, and the revision that produced it, are described in Section 4.5.

This audit does, however, group the classes along one axis, because that axis is the question being asked: **whether a study can support a claim about humans**. Systematic reviews, trials and observational studies are counted together as human evidence; animal and mechanistic work are counted together as pre-clinical. That grouping is an analytical choice made for this paper, not a hierarchy asserted by the resource, and Section 6 returns to its limits.

2.3 Structured extraction

Beyond bibliographic metadata, records carry curated fields for intervention, molecular target, species/model, and effect. A deeper extraction layer (sample size, effect size, stated limitations) is populated for **43 studies (11.1%)**, concentrated in the studies that carry the most argumentative weight: those cited by the gap layer and those with human or longevity endpoints. Coverage of this layer is incomplete by design rather than by oversight. It is manual work, and it was prioritised where precision matters most.

Identifier coverage is high: DOI 386/386 (100%), PMID 357/386 (92.5%), PMCID 203/386 (52.6%), abstract text 377/386 (97.7%).

2.4 Gap derivation

The gap layer is generated from the classified corpus by looking for three structural patterns: **evidence deserts** (a molecular node that is well characterised mechanistically but has no organismal or human phenotype test), **mechanism-to-outcome gaps** (a mechanism assumed to mediate an outcome but

never tested by blocking it), and **contradictions/tensions** (two well-supported findings that cannot both be straightforwardly true of the same intervention).

Each gap is stated as an explicit falsifiable hypothesis paired with a concrete proposed experiment, is linked to its constituent studies, and carries a subjective confidence weight. Gap statements are labelled as hypotheses throughout the resource and are not presented as findings.

2.5 Availability

The corpus, the derived data files and the code that generates the published resource are openly available (Section 7). Every number reported here was computed directly from the released data files and can be reproduced from them with the script included as supplementary material.

3. What the corpus contains

At the level of raw composition, the corpus divides as follows: **mechanistic and in-vitro work accounts for 198 studies (51.3%)**, animal studies for 107 (27.7%), human evidence for 39 (10.1%), and narrative reviews for 39 (10.1%).

By subject category, a separate axis from study system, the split is Mechanism 207 (53.6%), Animal 90 (23.3%), Review 40 (10.4%), Human 36 (9.3%), Negative result 6 (1.6%), side-effect studies 2, preprints 2, and one ongoing trial.

4. Findings

4.1 One study in ten speaks to humans

Combining the three human classes (systematic review, clinical trial, observational study) gives **39 of 386 studies, or 10.1%**. Animal and mechanistic work together account for **305 studies, or 79.0%**.

This is the headline number, and its interest lies less in the ratio itself — a mechanistically driven field is *expected* to be bottom-heavy — than in the fact that it is normally invisible. The ratio only becomes visible if someone labels every study by system and then counts. When the field is read paper by paper, or through reviews, the mechanistic base and the human tip are cited in the same register.

4.2 The human evidence is mostly about disease, not ageing

The 10.1% figure is itself generous with respect to ageing claims. Splitting the 25 human clinical trials by indication:

- **13 trials (52%) address licensed disease indications:** transplant immunosuppression (Kahan, 2000), renal cell carcinoma (Hudes et al., 2007; Motzer et al., 2008), pancreatic neuroendocrine tumours (Yao et al., 2011), hormone-receptor-positive breast cancer (Baselga et al., 2012), tuberous sclerosis complex and lymphangioleiomyomatosis (Bissler et al., 2008, 2013; Krueger et al., 2010; McCormack et al., 2011; Franz et al., 2013; Saxena et al., 2026), advanced solid tumours (Schram et al., 2025), and gastroenteropancreatic neuroendocrine tumours (Walter et al., 2026).

- **2 trials** are physiology experiments in healthy volunteers rather than therapeutic tests (Drummond et al., 2009, showing that rapamycin blocks the post-exercise rise in muscle protein synthesis; Lassila et al., 2026, on PXR-mediated liver growth through the AKT–mTOR axis).
- **10 trials** are ageing- or healthspan-oriented.

Within that last group, the outcome record reads as follows:

Trial	Design	n	Primary endpoint	Result
Mannick et al. 2014	RCT	—	Influenza vaccine response	Positive (~20% improvement)
Poulsen et al. 2013 *	RCT	24	Insulin sensitivity	Null
Romashkan et al. 2016 (CALERIE) *	RCT	218	Safety of 2-year caloric restriction	Safety only
Kraig et al. 2018	Pilot RCT	25	Safety / feasibility	Safety only
Mannick et al. 2018	Phase 2a RCT	264	Infection incidence	Positive
Chung et al. 2019	RCT (skin)	17	Senescence markers	Exploratory, positive
Mannick et al. 2021	Phase 2b/3 RCT	1024	Symptomatic respiratory illness	Missed (26% vs 25%, $p = 0.65$)
Moel et al. 2025 (PEARL)	RCT	114	Visceral fat (DXA)	Missed ($p = 0.942$)
Aware et al. 2026	Single-arm pilot	23	Cerebral blood flow	Positive in APOE4 carriers; no control arm
Singh et al. 2026 *	Crossover RCT	63	Autophagic flux (whole blood)	Null ($p = 0.28$)

* tests caloric restriction, resveratrol or dietary protein restriction rather than an mTOR inhibitor.

Sample size for Mannick et al. 2014 is not captured in the corpus record, an instance of the extraction-coverage limitation discussed in Section 5.

Read as a sequence rather than as ten separate results, this is the classic translational attrition curve. Every positive primary-endpoint result here comes from an early-phase, small, or uncontrolled design: a vaccine-response signal (2014), a phase-2a infection-incidence result in 264 people (2018), an exploratory 17-person skin trial (2019), and an uncontrolled 23-person pilot (2026). The one adequately powered confirmatory trial, the phase-3 continuation of that same phase-2a programme, by the same group, in 1024 participants, missed its primary endpoint. The only purpose-built healthspan RCT completed to date reported a pre-registered null. **No adequately powered controlled trial in this corpus has produced a confirmed positive primary-endpoint result for an ageing or healthspan outcome.**

The newest entry sharpens the point rather than softening it. Singh et al. (2026) ran a pre-registered randomized crossover trial in which 63 healthy adults completed four weeks each of average-protein and reduced-protein diets, with autophagic flux in whole blood as the primary outcome, measured by an assay built to capture flux rather than a static marker. Preclinical work predicts that protein restriction should inhibit mTORC1 and thereby raise autophagy. It did not: the adjusted mean difference was -8.46 ng LC3B-II/mg protein/h (95% CI -24.06 to 7.14 , $p = 0.28$), while metabolomics confirmed the diets had genuinely separated. This is the rarest kind of study in the whole corpus: a well-powered, pre-registered,

direct human test of a mechanism that the pre-clinical literature treats as established. It came back negative.

That sequence — deep mechanism, strong mouse data, an encouraging phase-2a, a failed phase-3, a null healthspan RCT, and now a null mechanistic test in humans — is legible in a single view only because the corpus labels and links the studies. It is precisely the pattern that disappears when the phase-2a result is cited on its own.

4.3 The human share is rising, slowly

Splitting the corpus by era shows a monotonic increase in the human share:

Era	Studies	Human	Share
before 2000	15	0	0.0%
2000–2009	91	5	5.5%
2010–2019	150	15	10.0%
2020 onwards	129	19	14.7%

The field is translating, but from a very low base: even in the most recent era, roughly six of every seven curated studies remain pre-clinical. This trend should be read with the curation bias in Section 5 in mind, since recent literature is over-sampled, but the *direction* is not an artefact of over-sampling, because over-sampling recent work would, if anything, inflate the mechanistic count as readily as the human one.

4.4 Negative results are 1.6% of the corpus

Six studies (1.6%) are classified as negative results, despite negative and contradictory findings having been sought deliberately during curation. They are worth naming, because each one constrains a claim that circulates freely: the phase-3 immune trial failure (Mannick et al., 2021), a null high-dose resveratrol RCT in obese men (Poulsen et al., 2013), a null supplement panel in mice (Strong et al., 2012), the ITP arm showing rapamycin but not resveratrol or simvastatin extends lifespan (Miller et al., 2011), a transient-rapamycin study whose positive lifespan effect came with sex-specific and toxicity caveats (Bitto et al., 2016), and a 2026 failure of rapamycin to protect spermatogonial cells from cisplatin damage (Jibrin et al., 2026).

That count is itself an undercount, and the corpus's own labelling is the reason. Singh et al. (2026) is a null result by any reasonable reading, but it is filed as a human trial rather than as a negative result, because the category field records what kind of study it is and not how it came out. Any corpus that separates those two things will undercount negatives on one of the two axes. A 1.6% formal rate in a corpus where negative results were actively hunted is an observation about the literature; the labelling gap is an observation about the corpus.

4.5 Classifying the evidence exposed a defect in the classification

The rubric originally used a lettered A–D scale. Under it, **237 studies (61.4% of the current corpus) fall into a single bucket, "D — Mechanistic/Review"**, which merged **198 primary mechanistic papers with 39 narrative reviews**.

In August 2026 the resource was described publicly on social media, and Brendan Manning (Harvard T.H. Chan School of Public Health) responded with a substantive objection: whether a mechanism has been described in humans should not be weighted, because rigorous and reproducible cell-biological and biochemical evidence shows how the core pathway functions regardless of whether it has yet been tested in organismal models. The objection was correct, and it identified a real defect rather than a presentational one. A lettered scale running A to D reads as a quality grade no matter how carefully the surrounding prose insists it is a design classifier, and placing a primary structural discovery in the same terminal bucket as a narrative review compounded the misreading.

The scheme was revised in September 2026. Letters were replaced with mnemonic codes for study type, narrative reviews were separated onto their own track rather than occupying the bottom rung, and the display layer was rebuilt so that nothing in the presentation implies a quality ranking. Internal identifiers were left unchanged, so no underlying record was re-graded: the revision changed what readers see, not what the data says.

Two things about this episode seem worth recording. First, the defect was invisible from inside the project and became visible only when an external specialist read the output, which is an argument for publishing rubrics rather than merely applying them. Second, the objection is not fully resolved by re-labelling, and this paper inherits the unresolved part. Section 2.2 groups the classes along a single axis — speaks to humans, or does not — because that is the question being asked here. That axis is defensible for this question and indefensible as a general ranking, which is exactly Manning's point. The honest general structure is two axes: strength of evidence for a claim about human outcomes, and strength of evidence for a claim about mechanism, scored separately, so that a definitive biochemical result is not implicitly demoted for being biochemical. That scheme is not implemented, and until it is, any single ordering of these classes should be read as a device for answering one question rather than as a statement about worth.

4.6 The gap layer

Ten gaps are currently derived from the corpus, distributed across four types: contradiction/tension (4), mechanism-to-outcome gap (3), evidence desert (2), and human-endpoint gap (1). Confidence weights range from 0.70 to 0.85. Collectively they cite **51 distinct studies (13.2% of the corpus)**, a mean of 6.6 per gap, with every cited identifier resolving to a corpus record.

Representative examples: whether autophagy is actually *required* for the mammalian lifespan benefit (assumed widely; shown directly only in fly and worm, plus one mouse over-expression study, with no study testing whether blocking autophagy abolishes the effect); whether mTORC1-selective, mTORC2-sparing inhibition can separate the longevity benefit from the insulin-resistance liability that is attributed to mTORC2 loss (Lamming et al., 2012); and pervasive, occasionally direction-flipping sex dimorphism in mTOR-longevity responses — S6K1 knockout extends median lifespan in female but not male mice (Selman et al., 2009), and rapamycin favours females at every dose tested (Miller et al., 2014) — a pattern that surfaced from extracted effect sizes across studies rather than from any individual paper's framing.

The first of those gained direct human relevance during the period covered by this revision. Singh et al. (2026) did not test whether autophagy is required for a lifespan benefit, but it did test whether a standard nutritional route to inhibiting mTORC1 raises autophagy in people at all, and found that it does not. A gap layer that flags the mechanism-to-outcome link as untested now has a human datum suggesting the first step of that link may not hold under the conditions usually assumed.

These are hypotheses, and are labelled as such. Their value is not that they are correct but that they are *derived and traceable*: each one names the studies that generate the tension and the experiment that would resolve it.

5. Limitations

Single-curator selection. The corpus was assembled by one curator. There is no second-rater agreement statistic for the class assignments, and none should be inferred. Inter-rater reliability on the rubric is the single most valuable missing measurement in this work, and is the obvious next study.

Not a systematic review. Selection was purposive, not exhaustive. The corpus cannot support prevalence claims about "the mTOR literature" as a whole; every proportion reported here is a proportion of *this corpus*.

Recency over-sampling. 103 of 386 studies (26.7%) carry a 2026 publication year, because curation began in 2026 and actively pulled current literature. Era-stratified figures (Section 4.3) are affected by this and should be read as directional.

A moving census. The corpus grows continuously. Version 1 of this paper reported 364 studies; this version reports 386, added over three days. Every figure here is a snapshot with a date on it, and the supplementary data file pins the exact snapshot the numbers came from.

Incomplete deep extraction. Sample size, effect size and stated limitations are populated for 43 studies (11.1%). Any analysis depending on those fields covers a small, non-random subset.

Classification is a judgement. Assigning a study to a class is rule-guided but not mechanical, particularly for mixed-model papers (human cells plus mouse, say). Section 4.5 documents one case in which the rubric itself was found wanting.

Indication classification is post hoc. The disease/physiology/ageing split in Section 4.2 was applied for this audit by reading each trial's stated indication and primary endpoint. It is not a pre-existing field in the corpus, and a different reader could reasonably reassign borderline cases.

6. Discussion

The finding that 10.1% of a curated mTOR corpus is human evidence is not, in isolation, a criticism of the field. Pathway biology is supposed to be bottom-heavy; mechanism precedes translation, and the mechanistic work in this corpus includes some of the most rigorous molecular biology of the last thirty years.

The finding matters because of what happens downstream. mTOR is unusual among ageing targets in that a drug acting on it is already licensed, already prescribed, and already taken off-label by people who are not transplant recipients or cancer patients. In that setting, the distance between "extends lifespan in mice" and "extends healthspan in humans" is not an academic distinction, and the evidence base actually supporting the second claim — ten trials, no confirmed positive primary endpoint, one failed phase-3, one pre-registered null on a healthspan outcome, and now one pre-registered null on the mechanism itself — is much thinner than the volume of surrounding literature suggests.

Making that composition visible is the contribution here. Not a new experiment; a census. The census is only possible because someone labelled 386 studies by system under a fixed rubric and published the result in a form that can be counted, queried and disagreed with.

The episode in Section 4.5 suggests the same logic applies reflexively. A classification scheme is itself a claim, and it improves under exactly the sort of external scrutiny that publishing it invites. The unresolved question it left behind — whether evidence strength is one axis or two — is, I think, the more interesting research problem of the two described in this paper, and this version of the paper states the single-axis grouping as the analytical device it is rather than dressing it as a hierarchy.

7. Data and code availability

- **Resource:** <https://mtor-atlas.org>
- **Archived dataset (DOI):** 10.5281/zenodo.22059963 (CC BY 4.0)
- **Source code:** <https://github.com/open-mtor-atlas/atlas> (code MIT; curated content and data CC BY 4.0)
- **Registry records:** [bio.tools olivers_mtor_atlas](https://bio.tools/olivers_mtor_atlas); FAIRsharing 8905; Wikidata Q141256074

All figures reported in this paper were computed directly from `studies_baked.json` and `gaps_baked.json` at the 15 September 2026 census. Both files, and a script that recomputes every number in a single run, are included as supplementary material.

8. Declarations

Funding. None.

Competing interests. The author built and maintains the resource audited in this paper. No financial competing interests.

Ethics. No human or animal subjects; this work analyses published literature only.

AI usage disclosure. Large language model assistance (Anthropic Claude) was used substantially throughout this project: in literature triage during curation, in drafting curated summary prose subsequently reviewed by the author, in generating candidate gap hypotheses from the classified corpus, in writing the code that produces the published resource, and in drafting this manuscript. Class assignments were made under the author's rubric with the author's review. Every numerical result reported here was computed programmatically from the released data files and re-verified against them; bibliographic details for all cited studies were checked against PubMed records rather than reproduced from memory. The author takes responsibility for the content.

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9. Changes in version 2

1. **Census updated** from 364 studies (12 September 2026) to 386 (15 September 2026). All figures recomputed; the headline human share moved from 10.4% to 10.1%.

2. **One new human trial** entered the corpus and is discussed in Sections 4.2, 4.4 and 4.6: Singh et al. (2026), a pre-registered crossover trial finding that dietary protein restriction does not raise autophagic flux in healthy adults.

3. **Section 2.2 reframed.** Version 1 presented the classes as a numbered hierarchy, which implied the quality ranking the resource explicitly disclaims. The classes are now presented as labels for the system a study was done in, with the human/pre-clinical grouping stated as an analytical choice made for this paper's question. Section 4.5 was extended to say what the unresolved two-axis problem means for this paper specifically.

4. **Section 4.4** now notes that the formal negative-result count undercounts, and why.

5. **A limitation added** on the moving census.

6. The study count was removed from the title, so that future versions do not need a title change.

References

Bibliographic details were verified against PubMed records.

Aware C, Neher CM, Woods C, et al. (2026). Rapamycin increases cerebral blood flow and modulates metabolic, inflammatory, and microbiome profiles in healthy middle-aged APOE4 carriers: a pilot single-arm trial. *Journal of Cerebral Blood Flow and Metabolism*. doi:10.1177/0271678X261490342

Baselga J, Campone M, Piccart M, et al. (2012). Everolimus in postmenopausal hormone-receptor-positive advanced breast cancer (BOLERO-2). *New England Journal of Medicine*. doi:10.1056/NEJMoa1109653

Bissler JJ, Franz DN, et al. (2008). Sirolimus for angiomyolipoma in tuberous sclerosis complex or lymphangioleiomyomatosis. *New England Journal of Medicine*. doi:10.1056/NEJMoa063564

Bissler JJ, Kingswood JC, Radzikowska E, et al. (2013). Everolimus for angiomyolipoma associated with tuberous sclerosis complex or sporadic lymphangioleiomyomatosis (EXIST-2). *The Lancet*. doi:10.1016/S0140-6736(12)61767-X

Bitto A, Kaeberlein M, et al. (2016). Transient rapamycin treatment can increase lifespan and healthspan in middle-aged mice. *eLife*. doi:10.7554/eLife.16351

Chung CL, Lawrence I, Hoffman M, et al. (2019). Topical rapamycin reduces markers of senescence and aging in human skin: an exploratory, prospective, randomized trial. *GeroScience*. doi:10.1007/s11357-019-00113-y

Drummond MJ, Rasmussen BB, et al. (2009). Rapamycin administration in humans blocks the contraction-induced increase in skeletal muscle protein synthesis. *Journal of Physiology*. doi:10.1113/jphysiol.2008.163816

Franz DN, Belousova E, Sparagana S, et al. (2013). Efficacy and safety of everolimus for subependymal giant cell astrocytomas associated with tuberous sclerosis complex (EXIST-1). *The Lancet*. doi:10.1016/S0140-6736(12)61134-9

Harrison DE, et al. (2009). Rapamycin fed late in life extends lifespan in genetically heterogeneous mice. *Nature*. doi:10.1038/nature08221

Hudes G, et al. (2007). Temsirolimus, interferon alfa, or both for advanced renal-cell carcinoma. *New England Journal of Medicine*. doi:10.1056/NEJMoa066838

- Jibrin A, Mitchell RT, Lopes F (2026). Rapamycin co-exposure fails to reduce cisplatin-induced damage in GC6-spg spermatogonial cell lines. *Reproduction & Fertility*. doi:10.1530/RAF-26-0044
- Kahan BD, for the Rapamune US Study Group (2000). Efficacy of sirolimus compared with azathioprine for reduction of acute renal allograft rejection. *The Lancet*. doi:10.1016/S0140-6736(00)02480-6
- Kraig E, Linehan LA, Liang H, et al. (2018). A randomized control trial to establish the feasibility and safety of rapamycin treatment in an older human cohort. *Experimental Gerontology*. doi:10.1016/j.exger.2017.12.026
- Krueger DA, et al. (2010). Everolimus for subependymal giant-cell astrocytomas in tuberous sclerosis. *New England Journal of Medicine*. doi:10.1056/NEJMoa1001671
- Lamming DW, Sabatini DM, et al. (2012). Rapamycin-induced insulin resistance is mediated by mTORC2 loss and uncoupled from longevity. *Science*. doi:10.1126/science.1215135
- Lassila P, Karpale M, Kumm O, et al. (2026). Pregnane X receptor-mediated liver growth: a controlled clinical trial in healthy volunteers. *Archives of Toxicology*. doi:10.1007/s00204-026-04526-5
- Mannick JB, Del Giudice G, Lattanzi M, et al. (2014). mTOR inhibition improves immune function in the elderly. *Science Translational Medicine*. doi:10.1126/scitranslmed.3009892
- Mannick JB, Morris M, Hockey HP, et al. (2018). TORC1 inhibition enhances immune function and reduces infections in the elderly. *Science Translational Medicine*. doi:10.1126/scitranslmed.aag1564
- Mannick JB, Teo G, Bernardo P, et al. (2021). Targeting the biology of ageing with mTOR inhibitors to improve immune function in older adults: phase 2b and phase 3 randomised trials. *Lancet Healthy Longevity*. doi:10.1016/S2666-7568(21)00062-3
- McCormack FX, Inoue Y, Moss J, et al. (2011). Efficacy and safety of sirolimus in lymphangioleiomyomatosis (MILES). *New England Journal of Medicine*. doi:10.1056/NEJMoa1100391
- Miller RA, Harrison DE, et al. (2011). Rapamycin, but not resveratrol or simvastatin, extends life span of genetically heterogeneous mice. *Journals of Gerontology Series A*. doi:10.1093/gerona/glq178
- Miller RA, Harrison DE, Astle CM, et al. (2014). Rapamycin-mediated lifespan increase in mice is dose and sex dependent and metabolically distinct from dietary restriction. *Aging Cell*. doi:10.1111/accel.12194
- Moel M, Harinath G, Lee V, et al. (2025). Influence of rapamycin on safety and healthspan metrics after one year: PEARL trial results. *Aging (Albany NY)*. doi:10.18632/aging.206235
- Motzer RJ, et al. (2008). Efficacy of everolimus in advanced renal cell carcinoma (RECORD-1). *The Lancet*. doi:10.1016/S0140-6736(08)61039-9
- Poulsen MM, et al. (2013). High-dose resveratrol supplementation in obese men: an investigator-initiated, randomized, placebo-controlled trial. *Diabetes*. doi:10.2337/db12-0975
- Romashkan SV, Ravussin E, et al. (2016). Safety of two-year caloric restriction in non-obese healthy individuals. *Oncotarget*. doi:10.18632/oncotarget.8093
- Sabatini DM, et al. (1994). RAFT1: a mammalian protein that binds to FKBP12 in a rapamycin-dependent fashion and is homologous to yeast TORs. *Cell*. doi:10.1016/0092-8674(94)90570-3
- Saxena A, Drew CJ, Cannings-John R, et al. (2026). Everolimus for the treatment of neuropsychological deficits in tuberous sclerosis complex. *Journal of Medical Genetics*. doi:10.1136/jmg-2026-111529

- Saxton RA, Wolfson RL, Sabatini DM, et al. (2015). Structural basis for leucine sensing by the Sestrin2-mTORC1 pathway. *Science*. doi:10.1126/science.aad2087
- Schram AM, Meyerowitz JG, et al. (2025). The bi-steric, mTORC1-selective inhibitor RMC-5552 in advanced solid tumors: a phase 1 trial. *Clinical Cancer Research*. doi:10.1158/1078-0432.CCR-25-2112
- Selman C, Withers DJ, et al. (2009). Ribosomal protein S6 kinase 1 signaling regulates mammalian life span. *Science*. doi:10.1126/science.1177221
- Singh S, Fourrier C, Hein LK, et al. (2026). Reduced dietary protein intake does not alter autophagy in human blood: a randomized crossover study in healthy adults. *Clinical Nutrition* 65:106778. doi:10.1016/j.clnu.2026.106778
- Strong R, Miller RA, Harrison DE, et al. (2012). Evaluation of resveratrol, green tea extract, curcumin, oxaloacetic acid, and medium-chain triglyceride oil on life span of genetically heterogeneous mice. *Journals of Gerontology Series A*. doi:10.1093/gerona/gls070
- Walter T, Jann H, Ansquer C, et al. (2026). [177Lu]Lu-edotreotide versus everolimus for gastroenteropancreatic neuroendocrine tumours (COMPETE): a phase 3 randomised trial. *The Lancet*. doi:10.1016/S0140-6736(26)00604-5
- Yao JC, Shah MH, Ito T, et al. (2011). Everolimus for advanced pancreatic neuroendocrine tumors (RADIANT-3). *New England Journal of Medicine*. doi:10.1056/NEJMoa1009290